RV Educational Institutions ®
RV College of Engineering ®

Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi

Approved by AICTE, New Delhi

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Academic year 2023-2024 (Odd Sem)

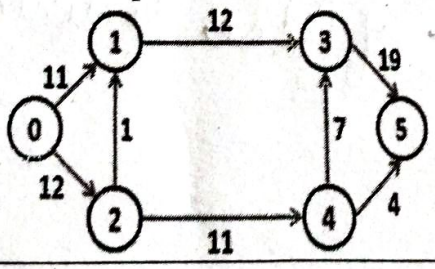
DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Date	February 2024	Maximum Marks	50
Course Code	21CS55B3	Duration	90 Min
Sem-V	Test-2	Staff: ASP/TP/KCG/KSN	

ADVANCED ALGORITHMS

(Common to CSE & ISE)

		Marks	BT	CO
1a.	Find all integers that leave remainders 1, 2, 3 when divided by 9, 8, 7 respectively using Chinese Remainder Theorem.	6	3	3
1b.	Prove that comparison-based sorting algorithm has a lower bound of (n log n).	4	2	1
2a.	Write an algorithm for radix sort and analyze its time complexity. Apply it to sort the elements COW,DOG,SEA,RUG,ROW,MOB,BOX,TAB,BAR.	6	3	4
2b.	Write RSA algorithm.	4	2	2
3a.	What is bipartite matching? With an example explain how maximum bipartite matching problems can be solved.	6	2	3
3b.	Solve the linear modular congruence relation $12x \equiv 6 \bmod 15$.	4	2	4
4.	Write the algorithm for counting sort and analyze its time complexity. Mention advantages and disadvantages of counting sort. Use the counting sort algorithm to sort the elements: 4, 6, 2, 4, 6, 2, 5, 3, 2, 0.	10	3	2
5.	Define (i) cut (ii) augmented path in a flow network. Apply Ford-Fulkerson method for the graph given below to find the augmented path and the maximum network flow. Draw the residual graph for each path and show all the calculations with steps: 	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test Max Marks	-	12	28	10	-	-	4	20	26	-

Course Outcomes:

1. Analyze various algorithms for their time and space complexity
2. Demonstrate a familiarity with major algorithms and data structures
3. Apply appropriate design techniques for solving real world problems.
4. Design and implement solutions using appropriate mathematical techniques

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